Publications

0.0.1 Refereed Scientific Articles

Journal Articles Zdeněk Zeman, Aki Mikkola, Grzegorz Orzechowski, Jiří Suder, and Milan Mihola. "Large language models in mechanical design of mechatronic systems: A review." *Advanced Engineering Informatics*. 2026. DOI

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Alexander Nemov, Aleksi Härkönen, Alfonso Garcia-Agundez, Qasim Khadim, Antti Ahola, Victor Zhidchenko, Dominik Walica, Emil Kurvinen, Heikki Handroos, Aki Mikkola, and Grzegorz Orzechowski. "Workflow for Real-Time Simulation-Based Fatigue Assessment in Mobile Machinery." *Advances in Mechanical Engineering*. 2026. DOI

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Seongji Han, Grzegorz Orzechowski, Jin-Gyun Kim, and Aki Mikkola. "Data-driven friction force

prediction model for hydraulic actuators using deep neural networks." *Mechanism and Machine Theory*. 2024. DOI

Arash Hashemi, Grzegorz Orzechowski, Aki Mikkola, and John McPhee. "Multibody Dynamics and Control Using Machine Learning." *Multibody System Dynamics*. 2023. DOI

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Ilya Kurinov, Grzegorz Orzechowski, Perttu Hämäläinen, and Aki Mikkola. "Automated Excavator Based on Reinforcement Learning and Multibody System Dynamics." *IEEE Access*. 2020. DOI

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and Andrzej Wróbel. “Brain-training for physical performance: a study of EEG-neurofeedback and alpha relaxation training in athletes.” *Acta Neurobiologiae Experimentalis*. 2015.

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0.0.2 Non-Refereed Scientific Articles

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Oleg Rogov, Peter Manzl, Johannes Gerstmayr, and Grzegorz Orzechowski. “Reinforcement Learning Methods for Multibody Systems Evaluated with Controlled Multi-Link Inverted Pendulum on the Cart.” *ECCOMAS Thematic Conference on Multibody Dynamics*. 2023.

Qasim Khadim, Grzegorz Orzechowski, Emil Kurvinen, and Aki Mikkola. “Numerical Optimization of Parameters Using the Covariance Matrix Adaptation Evolution Strategy in the Coupled Multibody Systems.” *ECCOMAS Thematic Conference on Multibody Dynamics*. 2023.

Seongji Han, Grzegorz Orzechowski, Jin-Gyun Kim, and Aki Mikkola. “A Data-Driven Metamodeling Technique for Friction Force in Hydraulic Actuators.” *Annual Meeting of the Korean Society of Mechanical Engineers*. 2023.

Grzegorz Orzechowski and Aki M. Mikkola. “High-fidelity Multibody Systems in Model-based Control Applications.” *IUTAM Symposium on Optimal Design and Control of Multibody Systems*. 2022.

Grzegorz Orzechowski, Perttu Hämäläinen, and Aki M. Mikkola. “Force Approximation of the Hand on the Climbing Wall Handle.” *ECCOMAS Thematic Conference on Multibody Dynamics*. 2019.

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Grzegorz Orzechowski and Janusz Frączek. “Modeling of the beams with taper cross section using absolute nodal coordinate formulation.” *19th International Conference on Computer Methods in Mechanics*. 2011.

Tomasz Krawczyk, Grzegorz Orzechowski, and Janusz Frączek. “Simulation studies of dynamics of the robot flexible power transmission (in Polish).” *10th Anniversary Scientific-Technical Conference, FEA programs in analysis, design and manufacturing*. 2007.

0.0.3 Scientific Books (Monographs)

Special Issue of a Journal Johannes Gerstmayr, Grzegorz Orzechowski, and Alessandro Tasora. “Special Issue on Artificial intelligence within multibody system dynamics.” *Multibody System Dynamics*. 2024. Link

Pierpaolo Belardinelli, Kiran X. D’Souza, Grzegorz Orzechowski, and Hiroyuki Sugiyama. “Special

Issue: Selected Papers From 2023 Multibody Systems, Nonlinear Dynamics, and Control Conference." *ASME Journal of Computational and Nonlinear Dynamics*. 2023. DOI

0.0.4 Theses

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